Florian **Franzen**

Leadership | Architecture | Infrastructure | Research

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Technical leader for complex infrastructure and protocol systems, from specification through implementation, deployment and operations. From setting technical strategy, research and roadmaps and making the architecture decisions to hiring for the team and driving partnerships and integrations, I will take systems from idea to production and keep them running there.

I have led engineering, research and infrastructure for Substrate-, Ethereum- and Solana-based networks through to mainnet launch. Today I provide technical leadership and engineering support to startups building protocols and infrastructure, through my studio F.R.E.D.

Behind that sits an unusually deep range: years in neuroscience and academic engineering, embedded hardware and prototype manufacturing, security and reverse engineering, data analysis and machine learning (including AI/LLMs), and full-stack web development. It lets me own a system from the circuit board to the production network.

Professional Experience

Current
Jun. 2026

Head of Research, Flowra, Remote
Partner with the CTO on the block engine: monitor performance, investigate deviations, review code changes and propose alternative approaches. Track competing block engines and developments across the Solana network. Publish regular articles on protocol changes and their impact on product strategy.
Agave | Alpenglow | Firedancer | Jito | MEV | Rust | Solana | TypeScript | Tokio

Apr. 2026
Dec. 2025

Senior Research Engineer, Raiku Labs Ltd, Remote
Design and implement auction marketplace. Interface with on-chain programs and data. Optimize performance, scheduling and blockspace utilization. Support DevOps with upcoming changes. Liaison for the Research team.
Anchor | Helm | Jito | Kubernetes | MEV | Rust | Solana | Tokio | Tonic

Dec. 2025
Aug. 2025

Head of Protocol Research, Gluwa, Remote
Direct research strategy and execution. Support product goals across technical teams. Lead evaluation of emerging technologies and advise the Chief Architect and Head of Engineering on critical technical decisions.
Cairo | Python | Rust | Solidity | Substrate | TypeScript

Aug. 2025
Dec. 2024

Head of Infrastructure, Analog, Remote
Oversee core blockchain infrastructure: mainnet launch, feature development, DevOps/SecOps, exchange and wallet integrations. Deliver the official Substrate-Ethereum bridge and containerize infrastructure for Kubernetes/Helm, optimizing reliability and cost. Select key hardware, run key ceremonies and define signing policies. Execute on-chain governance, fund management and runtime upgrades alongside founders.
Alloy | GraphQL | Hardhat | Helm | Kubernetes | PolkaVM | React | Rust | Solidity | Substrate | TypeScript | Tokenomics | WASM

Dec. 2024
Jun. 2023

Head of Research, Analog, Remote
Research and prototype core protocol components. Lead DevOps/SecOps and the technical side of product management, hiring and partner integrations. Report directly to founders.
Cryptography | Game Theory | Helm | Kubernetes | Protocol Design | Rust | Security | Substrate | Tokenomics | WASM

May 2023
Mar. 2020

Polkadot Specification Lead, Web3 Foundation, Zug, Switzerland
Co-develop the implementable Polkadot specification alongside the research team. Validate external implementations through a custom cross-implementation test suite. Shepherd the multi-client standardization process.
AsciiDoc | C++ | CMake | Go | Gossamer | Kagome | Julia | k-framework | Python | Ruby | Rust | Substrate | WASM

May 2019
Jul. 2016

Doctoral Researcher in Neuroinformatics, Research Center caesar, Bonn, Germany
Develop 3D environment and pose reconstruction from video data using non-linear optimization of multi-view geometric graphs. Build an ultra-lightweight tetherless recording platform for GPS, IMU, and barometric sensors. Reverse-engineer and extend a consumer-grade 360° camera. Develop advanced camera calibration, synchronization, and spatial tracking system.
Blender | C | C++ | CMake | Docker | IDA Pro | i²c | Jupyter | Matlab | Nix | OpenCV | Python | LSD-SLAM | SPI | Teensy | Qt

Jul. 2016
Nov. 2015

Network & System Administrator, Research Center caesar, Bonn, Germany
Install and configure wired and wireless infrastructure for 200+ clients, including firewalls, routing, switching, VLANs, 802.1x authentication, VPN, RADIUS, and SIP telephony.
Bash | Cisco ASA | Cisco IOS | Cisco ISE | Cisco WLC | Microsoft Active Directory

Freelance Experience

Current – 2014 | **Founder & Technical Lead, Franzen Research, Engineering & Development**
Technical leadership and engineering support for startups building protocols and infrastructure: architecture, protocol research, infrastructure and key management, security. Prototyping and embedded development, scientific computing, data digitization and electronics repair.
C/C++ · ESP32 · Key Management · KiCad · Kubernetes · Nix · Protocol Design · Rust · Smartcard · Solana · Substrate

Current | **Technical Advisor, Stealth Startup**
Evaluate specification and launch readiness for the initial data room: spec-to-implementation consistency, testnet and mainnet readiness (genesis ceremony, validator onboarding, operator documentation), governance and upgrade path, and audit readiness. Take part in founder meetings and technical investor conversations.
Intents · MEV-protection · Quantum-safety · Rust · Sharding

2019 – 2018 | **Software and Network Engineer, White Matter LLC**
Port a C++ API from Winsock to cross-platform Asio. Design binary network protocols, optimize packet throughput and stability, enforce thread safety, implement unit tests, and extend the Open Ephys plugin.
Asio · C++17 · Catch2 · CMake · Doxygen · fmtlib · JUCE · Linux · spdlog · Windows

2015 | **Software Engineer, Blackrock Microsystems LLC**
Add Blackrock file import and live data streaming to Neuroscope. Extend the Cerebus SDK, refactor and debug Neuroscope core, and establish CI/CD pipelines.
C++ · CMake · Qt · Blackrock SDK · Linux · Windows

Education

2014 | **Master of Neural & Behavioural Science, International Max Planck Research School, Tübingen, Germany**
Thesis: "Evaluation of a Statistical Model for Cortical State Identification"
Final grade: 2.1

2012 | **Bachelor of Cognitive Science, University of Tübingen, Tübingen, Germany**
Thesis: "Neural Circuits of Locomotor-related Response Modulation in Mouse Primary Visual Cortex"
Final grade: 2.1

Publications

2014 P. Putzky, **F. Franzen**, G. Bassetto, J. H. Macke. A Bayesian model for identifying hierarchically organised states in neural population activity. *Advances in Neural Information Processing Systems 27*. 2014.

2013 A. Vaiceliunaite, S. Erisken, **F. Franzen**, S. Katzner, L. Busse. Spatial integration in mouse primary visual cortex. *J Neurophysiol*. 2013 May 29.

Teaching

2023 *10 days* | **Polkadot Blockchain Academy, Universidad de Buenos Aires, Buenos Aires, Argentina**
Teaching cryptography, economics and blockchain

2022 *1 week* | **Polkadot Blockchain Academy, Corpus Christi College, Cambridge, England**
Teaching economics through interactive games

2015 *2 weeks* | **Open Lab Ware Course, TReND in Africa, Addis Ababa, Ethiopia**
Building and calibrating delta-style 3D printers, CAD, electrical circuit design, microcontroller programming